

A sixth personality domain that is independent of the Big Five domains: The psychometric properties of the HEXACO Personality Inventory in a Japanese sample

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Abstract: We investigated the reliability of the HEXACO-Personality Inventory (HEXACO-PI) using the NEO Five-Factor Inventory (NEO-FFI) and the International Personality Item Pool (IPIP) Big Five scales in a Japanese sample of 492 respondents (mean age = 20.4 years). The psychometric properties of the HEXACO-PI were examined and found to be satisfactory. The results of the factor analysis showed that the six hypothesized personality domains were found independently. A total of 307 participants responded to the NEO-FFI and the IPIP Big Five scales. The results showed that the five domains of the HEXACO model corresponded to the Five-Factor domains and were relative to the Big Five personality domains as a whole, but the sixth domain, Honesty-Humility, showed weak to moderate correlations with some factors of the NEO-FFI and the IPIP Big Five scales. These results suggest that the Honesty-Humility domain is independent of the five fundamental personality factors and that the HEXACO-PI is a useful and reliable questionnaire for investigating personality structure in cross-cultural and other context.

Key words: HEXACO model, HEXACO-PI, Big Five model, Five-Factor model.

One of the fundamental challenges of personality research is discovering the core structure of personality. For researchers who investigate personality, it is essential to determine the most reliable and stable dimensional model of personality because the fundamental framework of personality is required for research on personality.

Models of personality structure have been proposed and tested for many decades, but it was only in the mid-1980s that a consensus on personality structure began to emerge based on the results of a number of studies on the descriptive adjectives of personality (or “psycholexical” studies) (Goldberg, 1990; Saucier & Goldberg, 1996). Currently, many personality researchers think that a variety of human personality traits can be summarized in

terms of five independent dimensions (McCrae & John, 1992). These five dimensions were called the “Big Five” factors (Goldberg, 1990, 1993), and consist of: (a) Extraversion, (b) Emotional Stability (the reverse of Neuroticism), (c) Agreeableness, (d) Conscientiousness, and (e) Intellect/Imagination. These factors became a basis for personality research using a questionnaire by McCrae and Costa (1985), who modified and expanded the NEO personality model to include all the Big Five factors. They designated their personality structure model as the Five-Factor model, and it was very similar, but not quite identical to, the Big Five model. The dimensions in this novel model were named Extraversion, Neuroticism, Agreeableness, Conscientiousness, and Openness to Experience.

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Since the 1990s, a number of reliable and valid assessment tools have been developed based on the Big Five and the Five-Factor models. Costa and McCrae (1992) introduced the NEO Personality Inventory-Revised (NEO-PI-R) and its short version, the NEO Five-Factor Inventory (NEO-FFI). Goldberg (1992) introduced adjective scale markers for the Big Five and developed questionnaire scales with an identical structure using the International Personality Item Pool (IPIP; Goldberg, 1999). Goldberg (1999) constructed an additional set of scales assessing the Five-Factor Model dimensions, and these scales were approximately parallel to the NEO-PI-R/NEO-FFI scales. A significant amount of research has confirmed the psychometric quality of these instruments (e.g., Costa & McCrae, 1995; Goldberg, 1999).

The theory of personality described by five fundamental domains has been supported by research on personality descriptive adjectives in English (Goldberg, 1990; Saucier & Goldberg, 1996), other Western languages (Borkenau & Ostendorf, 1990; Caprara, Perugini, & Barbaranelli, 1994) and non-Western languages (Hahn, Lee, & Ashton, 1999). These results by cross-cultural or cross-language research support the validity of the Five-Factor (or the Big Five) model of personality.

It appeared that the issue of the fundamental personality dimensions was decided by the Five-Factor (or the Big Five) model, but since 2000 a number of lexical investigations in various languages have shown that six factors repeatedly emerge (Ashton et al., 2004) as dimensions of personality. The six-factor “HEXACO” model consists of five dimensions that are similar to those in the Five-Factor Model, with an additional “Honesty-Humility” dimension. This sixth factor measures individual differences in peoples’ sincerity, fairness, greed avoidance, and modesty that are not accounted for by the Big Five/Five Factor questionnaires. It has been shown to predict several personality traits such as workplace delinquency, risk-taking behavior, vengefulness, and creativity (Ashton & Lee, 2008; de Vries, de

Vries, & Feji, 2009; Johnson, Rowatt, & Petrini, 2011; Lee, Ashton, & de Vries, 2005; Silva, Kaufman, Reiter-Palmon, & Wigert, 2011). The name HEXACO derives from the initials of the six domains, Honesty-Humility (H), Emotionality (E), eXtraversion (X), Agreeableness (A), Conscientiousness (C), and Openness to Experience (O).

Lee and Ashton (2008) reported that a similar six-factor structure was found in twelve independent lexical studies of personality structure based on the results of rating an extensive number of adjectives that describe personality traits in the following languages: Croatian, Dutch, English, Filipino, French, German, Greek, Hungarian, Italian (in Rome and Trieste), Korean, Polish, and Turkish. Based on the results of this research, Lee and Ashton (2004) developed a questionnaire named the HEXACO Personality Inventory (HEXACO-PI) for measuring the six dimensions of personality based on the HEXACO model. Each domain of the HEXACO-PI consists of four facets based on the results of the lexical studies.²

The psychometric properties of the HEXACO-PI have been shown to be satisfactory (Lee & Ashton, 2004), and the six factors in the HEXACO-PI have been observed clearly by factor analysis in accordance with the HEXACO model (Lee & Ashton, 2004, 2006). Each factor of the HEXACO-PI except Honesty-Humility has shown a quite strong correlation with the corresponding factors of the Five-Factor model and a strong correlation with all the Big Five model factors except Emotional Stability and Agreeableness. As Ashton and Lee (2007) demonstrated, the HEXACO factors of Emotionality and Agreeableness are not identical to the similarly named factors in the Big Five model. Specifically, HEXACO Agreeableness hypothetically shows positive correlations with Emotional Stability and Agreeableness of the Big Five, and HEXACO Emotionality is expected to show a negative correlation with Emotional Stability and a

²Facet names for each domain of the HEXACO-PI are shown in Table 1.

positive correlation with Agreeableness of the Big Five (Ashton & Lee, 2007).

There is a question about whether the five- or six-factor model is more accurate in describing the fundamental personality structure.

In Japan, there have been several research studies regarding the Five-Factor/Big Five personality domains. The majority of questionnaire research studies confirmed the Five-Factor/Big Five dimensions of personality traits (for example, Kashiwagi & Wada, 1996). The results based on the psycholexical approach are not essentially consistent. Tsuji collected 13 198 trait terms from a Japanese dictionary, extracted 400 important and common trait terms, obtained eleven factors by factor analysis, and concluded that these factors appear to share few features with the Big Five model (Tsuji, 2001). Kashiwagi (2002) concluded that the Big Five dimensions of personality traits were confirmed in Japanese. It appears that the issue of the fundamental structure of personality has been almost decided in Japan, and that the personality traits found are very similar to those in the countries (languages) mentioned above. A question remains as to whether the sixth factor, "Honesty-Humility", is found independently of the Five-Factor/Big Five domains.

The predominant aim of this study was to examine the question of the independence of the "Honesty-Humility" domain using the HEXACO-PI. Because there is no substantial research with regard to the HEXACO model in Japan, except for some short reports about the model (Wakabayashi, 2009; Yasui & Tsuji, 2013), we examined the HEXACO-PI in terms of its psychometric properties and factor structures in a Japanese sample. The relations between the HEXACO-PI and the Five-Factor and the Big Five personality scales were investigated. It should be possible to determine in a Japanese sample whether the Honesty-Humility domain is a fundamental dimension of personality, as are the domains of the Five-Factor/Big Five models, as has been suggested by a number of lexical studies of personality structure, which rated an extensive number of adjectives that describe personality traits in different cultures (Lee & Ashton, 2008).

Method

Participants

A total of 492 undergraduate students (265 males and 227 females; mean age = 20.4 years, $SD = 1.45$ years) in three universities in Japan responded to the HEXACO-PI. Of the total, 307 (193 males and 114 females) of the students answered two personality inventories, the NEO-FFI and the IPIP Big Five scales. All the participants gave informed consent to participate in the study and received course credit for their participation.

Measures

Japanese HEXACO-PI. The original HEXACO-PI consists of 24 facet-level trait scales that are subsumed within six higher-order personality dimensions. Each scale consists of eight items, constituting a questionnaire of 192 items (Lee & Ashton, 2004). We used the Japanese version of the HEXACO-PI, which was translated using the following procedure. One of the authors of this paper translated the English-language HEXACO-PI items into Japanese. An English-Japanese bilingual psychologist checked the Japanese items to ascertain how well they corresponded to the original items. Native-English speakers who could understand Japanese translated the Japanese items into English, and then they were evaluated by the authors to ascertain how well they corresponded to the original English items.³ The final version of the Japanese HEXACO-PI features the identical five-point scale format as the English version.

The NEO-FFI. The NEO-FFI consists of 60 items based on the Five-Factor model (Costa & McCrae, 1992; the Japanese version, Shimonaka, Nakazato, Gondo, & Takayama, 1999).

The IPIP Big Five scales. A 50-item scale from the IPIP (Goldberg, 1999) was used to

³One of the authors of the original HEXACO-PI assisted in the design of the Japanese HEXACO-PI, in the translation and back translation of the items. We are much obliged for his assistance.

represent the lexical Big Five factors. These five IPIP scales were found to correlate strongly with the five factors derived from 100 unipolar Big Five adjective markers (Goldberg, 1992), ranging from $r = .66$ to $.90$. The items of the IPIP scales were translated into Japanese in the identical manner as was the HEXACO-PI.

The response format for all the measures described above was a five-point scale: 1 = *strongly disagree*, 2 = *disagree*, 3 = *neutral*, 4 = *agree*, and 5 = *strongly agree*.

Procedure

The Japanese version of the HEXACO-PI was administered in groups during the subjects' psychology classes. Approximately two-thirds ($N = 307$) of the participants were randomly selected to respond to the NEO-FFI and the IPIP Big Five scales, which were administered at a 2-week interval, approximately 1 month after they took the HEXACO-PI.

Results

Fundamental psychometric properties of the Japanese HEXACO-PI

All the questionnaires that included items with no response were rejected. The participants described above completed all the items and only data from these participants were included in the analyses.

The means and standard deviations of the six domains and the four facets in each domain for the entire sample, and for males and females separately, are shown in Table 1. There were significant sex differences in three domains (Honesty-Humility, Emotionality, and Extraversion); females scored higher than males on these domains. The considerable sex differences observed for Honesty-Humility and Emotionality are consistent with the results found in other cultural contexts (see Ashton & Lee, 2007). Kurtoses and skewnesses showed that the scores were normally distributed in each domain scale and its facets. As shown in Table 1, the internal consistency reliabilities (Cronbach's α) of the six domains ranged from $.84$ to $.90$ ($M = .87$), and those of the 24 facets

ranged from $.62$ to $.84$ ($M = .75$). Two facets of Emotionality showed slightly low internal consistencies compared with the other facets (Fearfulness = $.62$; Anxiety = $.64$). These facets contained a number of items that narrowly correlated with their facet scores. Regarding Fearfulness, the correlation between the item response for, "I would feel afraid if I had to travel in bad weather condition" and the facet score was $r = .17$. For Anxiety, the correlation between the item response for, "I tend to remain calm even when other people get stressed out" (reversal item) was $r = .07$. The item-facet correlations did not approach $r = .3$ for four items of Fearfulness and for two items of Anxiety.

Factor analyses

A principal factor analysis was applied to the responses to 192 items to investigate the factor structure of the Japanese HEXACO-PI. The first 10 eigenvalues were 12.8, 11.5, 9.4, 7.9, 6.9, 5.8, 4.1, 3.9, 3.4, and 2.8; the last important decrease in the eigenvalues occurred between the sixth and seventh factors. A plot of the eigenvalues suggested a six-factor solution would be optimum. The first six factors collectively explained 28.3 % of the total variance. These factors were rotated to a varimax solution. The results showed that the majority of the items loaded on their target factor and showed a six-factor structure, which corresponded to the HEXACO model. Factor I consisted of the items for Extraversion, and the other factors including the following: Factor II was Honesty-Humility, Factor III was Agreeableness, Factor IV was Emotionality, Factor V was Conscientiousness, and Factor VI was Openness to Experience. The scores of the 24 facet scales in the Japanese HEXACO-PI were submitted to a principal factor analysis. The first 10 eigenvalues were 3.5, 3.3, 2.6, 2.3, 1.9, 1.4, 0.9, 0.8, 0.7, and 0.6. A plot of the eigenvalues suggested that a six-factor solution was appropriate. The first six factors collectively explained 62.5 % of the total variance. The result of the varimax rotation showed a clear six-factor structure (Table 2). Each facet primarily loaded on the factor of the designated domain.

Table 1 Internal consistencies (Cronbach's α), means, and standard deviations of the HEXACO Personality Inventory scales in a Japanese sample

Scale (facet)	Cronbach's α	Total (<i>N</i> = 492)		Males (<i>n</i> = 265)		Females (<i>n</i> = 227)		Sex difference Cohen's <i>d</i>
		<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	
Honesty-Humility	.88	106.6	18.21	101.9	17.97	112.0	16.98	.57**
Sincerity	.71	22.4	5.43	21.7	5.39	23.1	5.39	.25**
Fairness	.79	29.7	6.77	27.8	7.08	31.9	5.55	.66**
Greed Avoidance	.81	25.8	6.60	24.7	6.87	27.1	6.01	.38**
Modesty	.79	28.7	5.86	27.8	5.98	29.7	5.56	.32**
Emotionality	.84	109.9	16.34	105.8	16.20	114.7	15.15	.57**
Fearfulness	.62	26.3	5.01	25.5	4.99	27.2	4.90	.33**
Anxiety	.64	28.5	5.35	27.9	5.36	29.3	5.25	.27**
Dependence	.82	26.9	6.54	25.7	6.61	28.4	6.15	.42**
Sentimentality	.73	28.1	5.95	26.7	5.92	29.8	5.51	.56**
Extraversion	.90	100.7	19.57	99.0	19.26	102.7	19.75	.19*
Expressiveness	.84	26.1	6.81	25.3	6.35	27.0	7.22	.24**
Social Boldness	.81	20.8	6.66	20.8	6.35	20.8	7.00	.01
Sociability	.78	28.3	5.95	28.0	6.09	28.7	5.76	.13
Liveliness	.79	25.5	6.24	25.0	6.09	26.2	6.36	.19*
Agreeableness	.88	91.5	17.87	90.3	16.31	92.8	19.45	.14
Forgiveness	.77	20.0	5.94	19.7	5.81	20.3	6.06	.11
Gentleness	.71	23.3	5.42	23.2	5.03	23.6	5.84	.07
Flexibility	.69	25.0	5.18	24.5	4.75	25.5	5.58	.19*
Patience	.78	23.2	6.45	22.9	6.29	23.4	6.63	.08
Conscientiousness	.87	95.2	17.85	95.3	18.57	95.1	16.99	.01
Organization	.83	22.2	7.26	22.4	7.38	22.0	7.12	.05
Diligence	.71	23.4	5.71	23.4	5.56	23.3	5.88	.01
Perfectionism	.70	26.7	5.61	26.5	5.96	26.9	5.15	.01
Prudence	.74	23.0	5.85	23.0	5.77	22.9	5.93	.08
Openness to Experience	.84	100.8	17.09	101.1	17.39	100.4	16.71	.04
Aesthetic Appreciation	.76	25.8	6.64	24.4	6.47	27.4	6.48	.45**
Inquisitiveness	.68	24.7	6.14	25.9	6.45	23.4	5.47	.42**
Creativity	.76	21.2	6.14	21.3	6.00	21.0	6.28	.05
Unconventionality	.68	29.1	5.20	29.5	5.02	28.6	5.36	.17

Note. Domains are shown in italic typeface. *t*-values for the difference between the means of males and females.

* $p < .05$, ** $p < .01$.

Intercorrelations among the domain scales

In the HEXACO model, independence among the six factors was hypothesized, similar to those in the Five-Factor/Big Five models. The correlations among the six factor-level scales are shown in Table 3. The correlations among the six domain scales were low in general, indicating that these domains were relatively independent of each other. The Honesty-Humility domain correlated slightly with Extraversion (negatively) and Agreeableness. Emotionality showed weak correlations with Extraversion,

Conscientiousness, and Openness (all negative). Extraversion correlated a little with Openness.

Convergent validities of the Japanese HEXACO-PI

The correlations among the Japanese HEXACO-PI scales and the NEO-FFI scales are shown in Table 4. Honesty-Humility was not strongly correlated with the NEO-FFI scales except for Agreeableness, with which it correlated slightly positively ($r = .25$) and Extraversion, with which it correlated slightly negatively ($r = -.19$). Medium to large positive

Table 2 Factor loadings of the 24 HEXACO Personality Inventory facet scales

Facet	Factor						Com.
	I	II	III	IV	V	VI	
Liveliness (X)	.811	.131	.041	.097	-.033	.088	.695
Social Boldness (X)	.710	-.039	.161	.132	-.131	.148	.588
Sociability (X)	.612	.220	-.377	-.009	-.157	-.105	.601
Expressiveness (X)	.583	-.274	-.256	-.247	-.066	.024	.546
Patience (A)	-.054	.788	.167	.071	.061	.084	.668
Gentleness (A)	.020	.778	.002	.080	.146	.054	.539
Flexibility (A)	-.041	.589	-.018	-.011	.170	-.233	.432
Forgiveness (A)	.164	.572	.081	-.157	.041	.109	.399
Dependence (E)	.129	-.003	-.673	-.268	-.105	-.128	.569
Anxiety (E)	-.100	-.173	-.644	-.060	-.051	.017	.461
Sentimentality (E)	.356	.094	-.641	.080	.121	.012	.568
Fearfulness (E)	-.207	-.175	-.436	-.006	.025	-.180	.297
Organization (C)	.108	-.038	.117	.674	.021	-.076	.487
Prudence (C)	-.134	.163	.177	.660	.056	-.177	.546
Perfectionism (C)	-.014	-.114	-.076	.602	.028	.114	.395
Diligence (C)	.398	.041	.072	.517	-.004	.171	.462
Greed Avoidance (H)	-.177	.018	.111	-.058	.794	.003	.678
Sincerity (H)	-.012	.129	.156	.062	.578	.163	.406
Fairness (H)	.109	.148	-.117	.290	.574	-.105	.472
Modesty (H)	-.249	.200	-.199	-.108	.568	-.131	.493
Aesthetic Appreciation (O)	.000	.056	-.115	.039	.136	.716	.549
Creativity (O)	.182	.025	.194	-.030	-.149	.566	.415
Inquisitiveness (O)	-.070	.024	.040	.074	-.008	.545	.310
Unconventionality (O)	.212	-.081	.137	-.166	.004	.488	.336

Note. *N* = 492. Loadings greater than .40 are shown in bold typeface. A = Agreeableness; C = Conscientiousness; E = Emotionality; H = Honesty-Humility; O = Openness to Experience; X = Extraversion.

Table 3 Intercorrelations among the scores of the six factor-level scales

Scale	1. Honesty-Humility	2. Emotionality	3. Extraversion	4. Agreeableness	5. Conscientiousness
2. Emotionality	-.02				
3. Extraversion	-.20**	.19**			
4. Agreeableness	.24**	-.10	.00		
5. Conscientiousness	.07	-.14**	.08	.03	
6. Openness to Experience	.04	-.15**	.13**	.03	.00

***p* < .01; **p* < .05.

correlations were found between the five domains of the HEXACO-PI and the NEO-FFI scales including the following: HEXACO Extraversion and NEO-FFI Extraversion ($r = .73$), HEXACO Conscientiousness and NEO-FFI Conscientiousness ($r = .68$), HEXACO Openness and NEO-FFI Openness ($r = .56$), HEXACO Agreeableness and NEO-FFI Agreeableness ($r = .52$), and HEXACO Emo-

tionality and NEO-FFI Neuroticism ($r = .49$). Three facets of Emotionality (Fearfulness, Dependence, and Sentimentality) in the HEXACO-PI did not correlate strongly with Neuroticism from the NEO-FFI, and two facets of Openness to Experience (Inquisitiveness and Creativity) from the HEXACO-PI showed medium to small correlations with Openness from the NEO-FFI (Table 4).

Table 4 Correlation among the HEXACO Personality Inventory and the NEO Five-Factor Inventory (NEO-FFI) scales

HEXACO scale	NEO-FFI scale				
	Neuroticism	Extraversion	Openness to Experience	Agreeableness	Conscientiousness
Honesty-Humility	-.01	-.19**	.07	.25**	.07
Sincerity	-.22**	-.04	-.01	.17**	.11
Fairness	-.09	.08	.01	.39**	.24**
Greed avoidance	-.03	-.24**	.05	.08	-.11
Modesty	.12*	-.15**	.01	.19**	-.17**
Emotionality	.49**	.17*	.02	.12*	-.12*
Fearfulness	.27**	-.11	-.05	.06	-.10
Anxiety	.58**	-.09	.09	-.08	-.11
Dependence	.37**	.24**	-.08	.03	-.18**
Sentimentality	.18**	.38**	.10	.31**	.05
Extraversion	-.15*	.73**	.08	.24**	.19**
Expressiveness	.15**	.40**	.07	-.01	-.06
Social Boldness	-.24**	.51**	.10	.13*	.30**
Sociability	-.04	.68**	.01	.37**	.12*
Liveliness	-.32**	.71**	.07	.30**	.25**
Agreeableness	-.23**	.33**	-.01	.52**	.02
Forgiveness	-.16**	.30**	.00	.22**	-.07
Gentleness	-.21**	.31**	.05	.51**	.10
Flexibility	-.11**	.19**	-.13*	.46**	-.03
Patience	-.23**	.22**	.02	.44**	.07
Conscientiousness	-.23**	.11*	.07	.18**	.68**
Organization	-.20**	.09	.02	.11	.57**
Diligence	-.22**	.29**	.13*	.14*	.56**
Perfectionism	.01	-.03	.11*	.05	.42**
Prudence	-.27**	-.01	-.02	.24**	.45**
Openness to Experience	-.10	.09	.56**	-.06	.04
Aesthetic appreciation	-.02	.04	.51**	.00	.02
Inquisitiveness	-.11	-.06	.36**	-.01	.09
Creativity	-.13*	.17**	.28**	-.05	-.08
Unconventionality	-.02	.08	.40**	-.14*	.04

Note. $N = 307$. Absolute values greater than .40 are shown in bold typeface. Domains are shown in italic typeface.

* $p < .05$; ** $p < .01$.

The correlations among the Japanese HEXACO-PI scales and the IPIP Big Five scales are shown in Table 5. On average, the results of the correlation analyses were similar to the results for the relation between the HEXACO-PI and the NEO-FFI. Honesty-Humility was relatively independent in the IPIP Big Five scales, although it showed weak correlations ($r = -.18$ to $.14$) with the five domains. The strongest correlation was found with Intellect/Imagination ($r = -.18$), with which Modesty correlated negatively at the facet level ($r = -.26$).

Two of the five domain scales of the HEXACO-PI showed a large positive correlation with the theoretical corresponding factor of the IPIP Big Five scales. HEXACO Extraversion correlated with IPIP Extraversion ($r = .78$) and HEXACO Conscientiousness correlated with IPIP Conscientiousness ($r = .76$). The remaining three domain scales of the HEXACO-PI showed only moderate correlation with their corresponding factors in the IPIP scales; HEXACO Emotionality correlated negatively with IPIP Emotional stability ($r = -.50$), HEXACO Agreeableness correlated

Table 5 Correlation among the HEXACO Personality Inventory and the International Personality Item Pool (IPIP) Big Five scales

HEXACO scale	IPIP scale				
	Emotional stability	Extraversion	Agreeableness	Conscientiousness	Intellect/Imagination
Honesty-Humility	.14*	-.15**	.14*	.15**	-.18**
Sincerity	.25**	-.01	.09	.17**	-.06
Fairness	.10	.04	.29**	.28**	-.05
Greed avoidance	.08	-.25**	-.03	.04	-.17**
Modesty	.00	-.21**	.06	-.03	-.26**
Emotionality	-.50**	.14*	.19**	-.09	-.07
Fearfulness	-.28**	-.10	.05	-.02	-.17**
Anxiety	-.57**	-.05	.03	-.04	-.01
Dependence	-.36**	.23**	.05	-.23**	-.07
Sentimentality	-.20**	.27**	.40**	.04	.06
Extraversion	.04	.78**	.26**	.03	.29**
Expressiveness	-.26**	.50**	-.01	-.17**	.21**
Social Boldness	.12*	.69**	.16**	.15**	.37**
Sociability	.05	.61**	.40**	.01	.08
Liveliness	.24**	.62**	.27**	.11	.22**
Agreeableness	.40**	.14*	.47**	.05	-.04
Forgiveness	.25**	.20**	.24**	-.10	.10
Gentleness	.34**	.07	.49**	.07	-.05
Flexibility	.23**	.06	.36**	.08	-.16**
Patience	.42**	.08	.38**	.10	-.04
Conscientiousness	.22**	.09	.26**	.76**	.09
Organization	.18**	.07	.14*	.70**	.00
Diligence	.14*	.28**	.21**	.47**	.15**
Perfectionism	-.02	-.02	.16**	.52**	.14*
Prudence	.35**	-.05	.27**	.53**	.01
Openness to Experience	.05	.08	.05	-.04	.49**
Aesthetic appreciation	-.03	.03	.08	-.04	.29**
Inquisitiveness	.12*	-.06	.05	.06	.27**
Creativity	.09	.17**	.05	-.02	.51**
Unconventionality	-.04	.10	-.07	-.13*	.31**

Note. $N = 307$. Absolute values greater than .40 are in bold typeface. Domains are shown in italic typeface. * $p < .05$; ** $p < .01$.

positively with IPIP Agreeableness ($r = .47$), and Openness correlated positively with IPIP Intellect/Imagination ($r = .49$). Agreeableness on the HEXACO-PI showed a moderate correlation with IPIP Emotional stability ($r = .40$).

Congruence of the factor loadings of the Japanese and the original HEXACO-PI

We calculated the congruence coefficients of the Japanese HEXACO factor loading and the factor loadings reported by Lee and Ashton (2004), which were as follows: Honesty-Humility = .96, Emotionality = .95, Extraversion = .93, Agreeableness = .95, Conscientiousness = .93, and

Openness to Experience = .94. These numbers (all above .90) suggested a strong similarity between the two models.⁴

Discussion

This study aimed to investigate the construct validity of the HEXACO-PI scales in a Japanese

⁴The calculations of the congruence coefficients of the Japanese HEXACO factor loading and the factor loadings reported in Lee and Ashton (2004) were performed by one of the authors of the original HEXACO-PI.

sample and to examine the relation between the factors of the HEXACO-PI and those of the NEO-FFI and the Big Five in Japan. In general, the results supported the cross-cultural stability of the HEXACO-PI.

The psychometric properties of the Japanese HEXACO-PI appeared to be nearly satisfactory. All six domain-level scales and the majority of the facet-level scales showed acceptable internal consistencies. Only two facets of Emotionality showed medium internal consistency. Those two facets, Fearfulness and Anxiety, contained a number of items that did not correlate with the facet scores. The majority of these items showed deviate response distributions. For Fearfulness, the response rates of *agree* or *strongly agree* exceeded 75% for three of eight items (e.g., for the item of, "People say that I am a fearless person," 82.6% of the participants answered *disagree* or *strongly disagree*). For Anxiety, less than 15% of the participants agreed (or strongly agreed) with two of eight items (e.g., for the item, "I get very anxious when waiting to hear about an important decision," only 10.1 % of the participants answered *agree* or *strongly agree*). In the former case, the answer *disagree* to the item might not necessarily imply, "I am a fearful person" for the majority of the participants. It might merely indicate, "I am not *so* fearless a person". In the latter case, a considerable number of participants who did not *agree* or *strongly agree* might imply that they would "get anxious to some extent (but not so much) when waiting to hear about an important decision." In this study, we translated original English items into Japanese using the back-translation process as mentioned previously. For a number of items, the Japanese description might imply a response that is excessively strong or weak compared with the English description. It would be better to adjust the descriptions in some items to avoid the deviating response rate, which might decrease internal consistency. However, because the number of those questionable items was quite small, it is possible to conclude that in general the Japanese HEXACO-PI is a parallel version of the original HEXACO-PI.

Sex differences were definitely found in the domains of Honesty-Humility and Emotional-

ity, and to some extent in the Extraversion domain; females scored higher than males on these scales. With regard to the sex differences these results were consistent with the results obtained in North American samples (Lee & Ashton, 2004, 2006).

The results of the factor analysis showed an obvious six-factor structure, particularly in the results from the facet level analysis, which corresponded to the structure hypothesized in the HEXACO model. This result suggested that the HEXACO-PI was factorially valid in this Japanese sample. The intercorrelation among the six domains did not show any clear relation except in the Honesty-Humility domain, which showed slight correlations with Extraversion ($r = -.20$) and Agreeableness ($r = .24$). The correlations on the facet level between Honesty-Humility and Extraversion showed that there were a number of mild correlations between Modesty (H) and Social Boldness (X) ($r = -.35$) and between Greed Avoidance (H) and Sociability (X) ($r = -.29$), although there was approximately no substantial correlation between the other facets for Honesty-Humility and Extraversion. At the facet level, there was a mild correlation between Modesty (H) and Flexibility (A) ($r = .29$). These results suggested the possibility that the Honesty-Humility domain is less independent than the other five domains that correspond to the Five Factor model.

Among the other five domains, the correlation between Emotionality and Extraversion was $r = .19$, which was not particularly high but was significant. This result was different from the findings in the original study (Lee & Ashton, 2004). At the facet level for Emotionality and Extraversion, a number of mild to moderate correlations were found including the following: the correlation between Sentimentality (E) and Sociability (X) was $r = .45$, between Sentimentality and Expressiveness (X) was $r = .30$, between Dependence (E) and Sociability was $r = .41$, and between Dependence and Expressiveness was $r = .31$. These results suggested that some aspects of Sentimentality and Dependence in Emotionality were connected with aspects of Sociability and

Expressiveness in Extraversion to some extent in the Japanese sample, while those aspects of Emotionality and Extraversion are independent of one another in the Canadian sample.

The Japanese HEXACO-PI domains showed relatively high convergent validities with the NEO-FFI scales and the IPIP Big Five scales. Regarding the relation with the Five-Factor model, the domains of the HEXACO-PI except Honesty-Humility corresponded to their counterpart in the NEO-FFI scales (Lee & Ashton, 2004). The results of the correlation analyses showed that two of the domain level traits (Extraversion and Conscientiousness) and their associated facets in the Japanese HEXACO-PI strongly correlated with their corresponding factors in the NEO-FFI. The remaining three domains (Emotionality, Agreeableness, and Openness to Experience) of the Japanese HEXACO-PI showed medium correlation with their corresponding factors in the NEO-FFI, and a number of the associated facets in these three domains and their corresponding factors in the NEO-FFI showed relatively weaker correlations than expected. As seen in Table 4, two facets of Emotionality showed mild to weak correlations with Neuroticism in the NEO-FFI ($r = .27$ for Fearfulness and $r = .18$ for Sentimentality). These results suggested that Emotionality in the HEXACO-PI is not identical to Neuroticism in the NEO-FFI for Japanese participants. Because Sentimentality showed mild correlations with Extraversion ($r = .38$) and Agreeableness ($r = .31$) rather than Neuroticism ($r = .18$) in the NEO-FFI, a number of items in this facet might be accepted as aspects related to empathy rather than sentimentality in the Japanese sample. The facet of Forgiveness for Agreeableness showed a weak correlation with Agreeableness for the NEO-FFI ($r = .22$). A number of items in this facet seem to relate to sociability rather than to forgiveness. In Openness, no facet showed a low correlation with Openness in the NEO-FFI, but the facet of Creativity showed only a mild correlation with it ($r = .28$).

These results suggested that the five domain level traits of the Japanese HEXACO-PI,

except Honesty-Humility, corresponded relatively to their counterparts in the Five-Factor scales as a whole, although some of the facets for Emotionality, Agreeableness, and Openness to Experience were partially different from their corresponding factors in the NEO-FFI.

With regard to the relationship with the Big Five model, Lee and Ashton (2004) hypothesized that HEXACO Extraversion and Conscientiousness were approximately parallel to their lexical Big Five counterparts. The results showed that high convergent validities were found for Extraversion ($r = .78$) and Conscientiousness ($r = .76$).

Emotionality and Agreeableness of the HEXACO model were not exactly identical to the similarly named factors in the Big Five model (Ashton & Lee, 2007). Lee, Ogunfowora, & Ashton (2005) suggested that HEXACO-PI Agreeableness and Emotionality constituted rotational variants of Big Five Agreeableness and Emotional Stability. These two domains (Emotionality and Agreeableness) of the Japanese HEXACO-PI correlated moderately with their counterparts in the IPIP Big Five scales (negatively with Emotional Stability and positively with Agreeableness), some of the facets of Emotionality and Agreeableness showed relatively small to medium correlations with the IPIP Emotional Stability and Agreeableness scales, respectively. For example, two facets in Emotionality correlated less than $r = .3$ with Emotional Stability in the IPIP Big Five scales (negatively, $r = -.28$ in Fearfulness and $r = -.20$ in Sentimentality). Similar to the result observed in the relation between the HEXACO-PI and NEO-FFI, it suggests that Sentimentality was not considered an aspect of emotional instability for Japanese participants. With regard to the correspondence in Agreeableness, Forgiveness (in Agreeableness) in the HEXACO-PI showed a low correlation with Agreeableness in the IPIP Big Five scales ($r = .24$), and the other two facets showed only mild correlations with Big Five Agreeableness. A number of aspects of Forgiveness might not be taken as a part of Agreeableness in the Japanese sample. Regarding Openness to Experience, HEXACO Openness showed only

a medium correlation with the IPIP Intellect/Imagination ($r = .49$), reflecting that HEXACO Openness excluded items involving the self-rating of intelligence that were represented in the IPIP Intellect/Imagination scale. Three of the four facets showed correlations of approximately $r = .3$ with Intellect/Imagination in the IPIP Big Five scales. Apparently, Openness in the HEXACO-PI is not an identical construct to Intellect/Imagination in the Big Five scales.

These findings suggest that the results involving the NEO-FFI are slightly different from the results involving the IPIP scales. This difference could be because of the different features associated with the two inventories. The NEO-FFI is based on the NEO-PI-R, which modified and expanded the NEO-PI to include all of the Big Five factors of Costa and McCrae (1992). In contrast, the IPIP Big Five factors are based directly on classic lexical findings.

The patterns of intercorrelations between the hypothetical counterpart domains of the HEXACO-PI scales and the NEO-FFI scales were approximately comparable to those of the HEXACO-PI and the IPIP Big Five scales. These results suggest that the Extraversion and Conscientiousness scales in the Japanese HEXACO-PI correspond to those in the NEO-FFI or those in the Big Five scales, but the other three domains in the Japanese HEXACO-PI do not absolutely correspond to the hypothetical counterpart domains in the NEO-FFI or to those in the Big Five scales.

Despite the results that the correlations between the corresponding domains were not necessarily sufficiently strong in the three domains of the HEXACO-PI and the Five-Factor/Big Five, the factor pattern of the Japanese HEXACO-PI replicated those obtained in the original study. The congruence coefficients of the Japanese HEXACO factor loading and the factor loadings of the Canadian sample (Lee & Ashton, 2004) showed a strong similarity ($M = 0.94$) between the two solutions. This result suggests that the HEXACO-PI and its six-factor structure are stable cross-culturally.

The Honesty-Humility domain was nearly independent of the other five domains of the

Japanese HEXACO-PI, as well as of the five fundamental personality factors of the NEO-FFI and the IPIP Big Five scales. These results suggest that the Honesty-Humility domain is different from the common personality dimensions described by the Five-Factor and the Big Five model. These results suggest that the Honesty-Humility domain is a valid new personality dimension independent of the Five-Factor (or Big Five) dimensions.

The results in this study suggest that the HEXACO-PI is a substantially reliable and valid instrument for personality description based on the HEXACO model, created by adding the Honesty-Humility domain to the Five-Factor model. Ashton and Lee (2005) reported that the NEO-PI-R contains certain Honesty-Humility-related content in its Agreeableness domain, and the Honesty-Humility scale showed the highest correlation with Agreeableness among the NEO-FFI scales. For the facet level scales, only the Fairness scale correlated with Agreeableness in this study. In general, the correlations between the Honesty-Humility domain and its four facets and the Five-Factor/Big Five scales were not very strong in the Japanese sample, supporting the hypothesis that the Five-Factor and Big Five model do not adequately capture the variance related to the Honesty-Humility factor.

The contents of the facets in the Honesty-Humility domain might be more desirable in the majority of social contexts than are those included in the other five personality domains. This hypothesis introduces the possibility that socially desirable responding would affect the scores on the Honesty-Humility scale more than it would for the other personality domains. Regarding this question, the other Five-Factor domains contain a certain number of socially desirable items to some extent as well, and the analysis indicated that the scores on the Honesty-Humility scale were no more negatively skewed than were the scores on the other five personality scales (the skewness of the scale of Honesty-Humility was -0.195). This result suggests that the responses to the items in this domain were not affected by social desirability.

Lee and Ashton (2006) reported that the self/observer convergent correlation of the Honesty-Humility scale was approximately the same as those of the other domain scales. This result suggests that scores on the Honesty-Humility scale are highly saturated with true substantive personality variance rather than with response style variance (e.g., social desirability).

There are a number of limitations in this study. First, the results are exclusively from the responses to questionnaires. It is possible that different results may emerge from analyzing the data on the Japanese trait terms from lexical research. The validity of the HEXACO model in Japan remains unclear, and there is a possibility that the Japanese lexical study does not support the six-factor model (Kashiwagi, 2002; Kashiwagi, Tsuji, Fujishima, & Yamada, 2005; Tsuji, 2001). Further research is needed to confirm whether the six-factor HEXACO model emerges from Japanese trait terms based on the psycholexical approach.

Second, the IPIP Big Five scales used in this study have not received sufficient examination, although the identical process of translation and back translation was used in the creation of the HEXACO-PI. The results suggest that the IPIP Big Five scales could be parallel to the Big Five domains based on lexical research, but it would be necessary to examine the reliability and validity of the IPIP Big Five scales.

Finally, the result that the reliability of the Japanese HEXACO-PI is confirmed does not necessarily mean that the HEXACO model is better than the Five-Factor/Big Five model for describing the fundamental personality dimensions in Japan. The HEXACO model consists of the Five-Factor domains with the Honesty-Humility domain, suggesting that the Five-Factor domains are a major part of personality structure. In other words, the HEXACO model may be a variant of the Five-Factor/Big Five model. The issue in this study is whether the sixth domain of Honesty-Humility is independent of the Five-Factor domains in the Japanese sample, and the result obtained from the HEXACO-PI nearly suggested the independence of Honesty-Humility.

Conclusion

The HEXACO model of personality has been proposed as a competitor to the widely accepted Five-Factor/Big Five model. In this study, as a first stage in the investigation of the reliability and validity of this model, we examined, in a Japanese context, the reliability of the HEXACO-PI, which is designed to measure six major dimensions of personality and their constituent facets. The results showed that the HEXACO-PI was a reliable and useful instrument to assess individual differences in the major personality dimensions in Japanese young adults. Based on the results of the factor analysis and item analysis, we recommend that a number of items should be revised to improve the reliability of the HEXACO-PI.⁵

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⁵The HEXACO-PI Japanese version is available from the HEXACO-PI-R website (hexaco.org) by K. Lee and M. C. Ashton.

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